

10. Clock Frequency Accuracy Measurement Circuit (CAC)

10.1 Overview

The clock frequency accuracy measurement circuit (CAC) counts pulses of the clock to be measured (measurement target clock) within the time generated by the clock to be used as a measurement reference (measurement reference clock), and determines the accuracy depending on whether the number of pulses is within the allowable range.

When measurement is completed or the number of pulses within the time generated by the measurement reference clock is not within the allowable range, an interrupt request is generated.

Table 10.1 lists the specifications of the CAC and Figure 10.1 shows a block diagram of the CAC.

Table 10.1 CAC Specifications

Item	Description
Measurement target clocks	The frequency of the following clocks can be measured. • Main clock • Sub-clock • HOCO clock • LOCO clock • IWDT-dedicated clock (IWDTCCLK) • Peripheral module clock B (PCLKB) • USB clock (UCLK) • External clock for the Ethernet-PHY (CLKOUT25M)
Measurement reference clocks	• External clock input to the CACREF pin • Main clock • Sub-clock • HOCO clock • LOCO clock • IWDT-dedicated clock (IWDTCCLK) • Peripheral module clock B (PCLKB) • USB clock (UCLK) • External clock for the Ethernet-PHY (CLKOUT25M)
Selectable function	Digital filter function
Interrupt sources	• Measurement end interrupt • Frequency error interrupt • Overflow interrupt
Low power consumption function	Module stop state can be set.

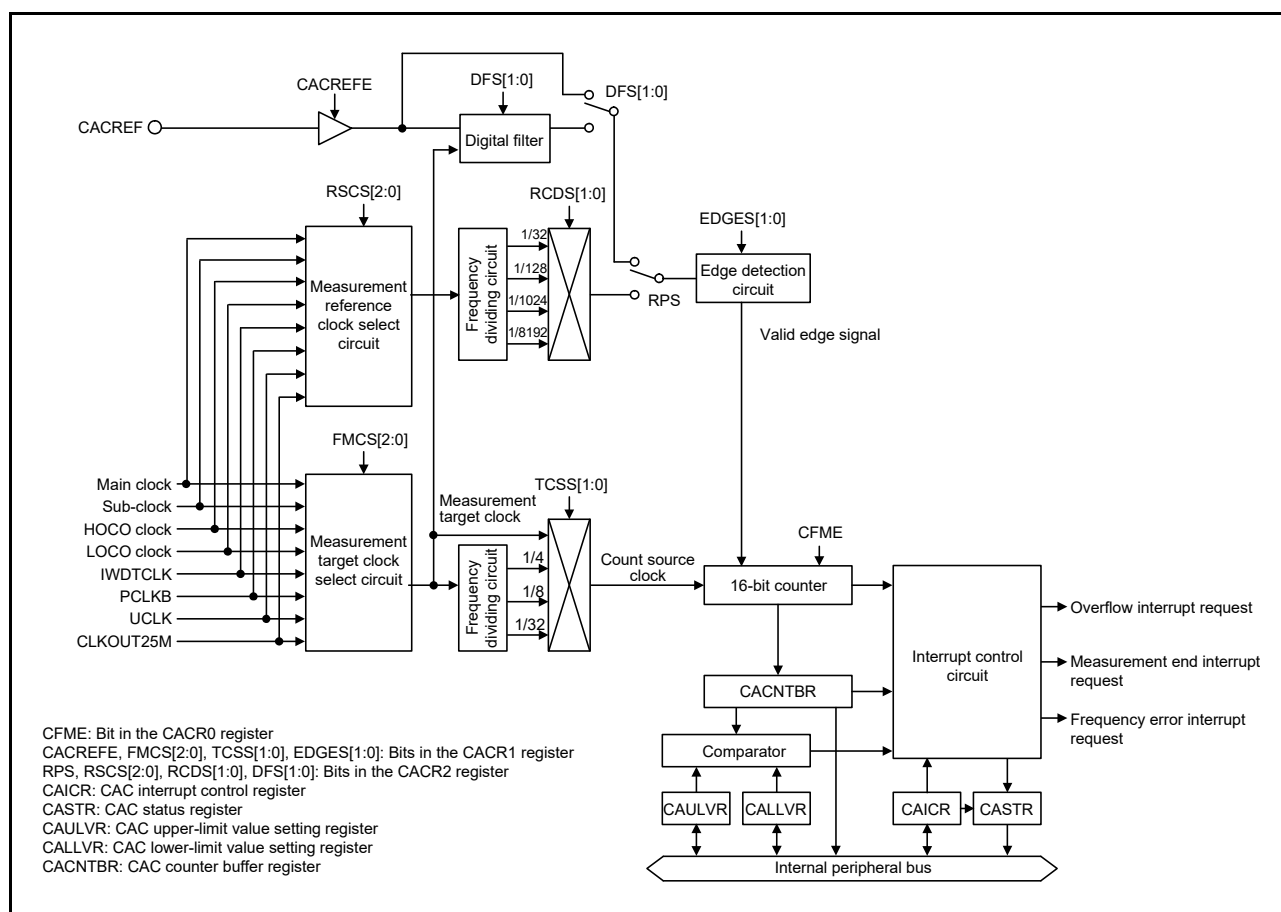


Figure 10.1 CAC Block Diagram

Table 10.2 shows the pin configuration of the CAC.

Table 10.2 Pin Configuration of CAC

Pin Name	I/O	Function
CACREF	Input	Measurement reference clock input pin

10.2 Register Descriptions

10.2.1 CAC Control Register 0 (CACR0)

Address(es): CAC.CACR0 0008 B000h

	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	CFME
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	CFME	Clock Frequency Measurement Enable	0: Clock frequency measurement is disabled. 1: Clock frequency measurement is enabled.	R/W
b7 to b1	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

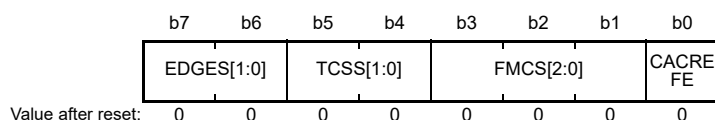
CFME Bit (Clock Frequency Measurement Enable)

This bit specifies whether clock frequency measurement is enabled or disabled.

When rewriting this bit, more time is required than other bits for the new value to be reflected in the register. Further write access to this bit are ignored until the current write access is reflected in the register. Read the bit to confirm that the rewrite has been reflected in the register.

10.2.2 CAC Control Register 1 (CACR1)

Address(es): CAC.CACR1 0008 B001h



Bit	Symbol	Bit Name	Description	R/W
b0	CACREFE	CACREF Pin Input Enable	0: CACREF pin input is disabled. 1: CACREF pin input is enabled.	R/W
b3 to b1	FMCS[2:0]	Measurement Target Clock Select	b3 b1 0 0 0: Main clock 0 0 1: Sub-clock 0 1 0: HOCO clock 0 1 1: LOCO clock 1 0 0: IWDTC-dedicated clock (IWDTCCLK) 1 0 1: Peripheral module clock B (PCLKB) 1 1 0: USB clock (UCLK) 1 1 1: External clock for the Ethernet-PHY (CLKOUT25M)	R/W
b5, b4	TCSS[1:0]	Timer Count Clock Source Select	b5 b4 0 0: No division 0 1: $\times 1/4$ clock 1 0: $\times 1/8$ clock 1 1: $\times 1/32$ clock	R/W
b7, b6	EDGES[1:0]	Valid Edge Select	b7 b6 0 0: Rising edge 0 1: Falling edge 1 0: Both rising and falling edges 1 1: Setting prohibited	R/W

Note: Set the CACR1 register when the CACR0.CFME bit is 0.

CACREFE Bit (CACREF Pin Input Enable)

This bit specifies whether the CACREF pin input is enabled or disabled.

FMCS[2:0]Bits (Measurement Target Clock Select)

These bits select the measurement target clock whose frequency is to be measured.

TCSS[1:0] Bits (Timer Count Clock Source Select)

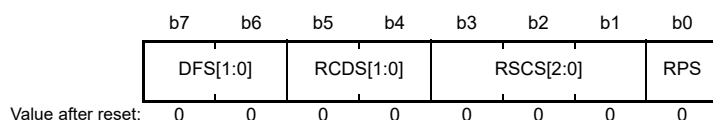
These bits select the count clock source for the clock frequency accuracy measurement circuit.

EDGES[1:0]Bits (Valid Edge Select)

These bits select the valid edge for the reference signal.

10.2.3 CAC Control Register 2 (CACR2)

Address(es): CAC.CACR2 0008 B002h



Bit	Symbol	Bit Name	Description	R/W
b0	RPS	Reference Signal Select	0: CACREF pin input 1: Internal clock (internally generated signal)	R/W
b3 to b1	RSCS[2:0]	Measurement Reference Clock Select	b3 b1 0 0 0: Main clock 0 0 1: Sub-clock 0 1 0: HOCO clock 0 1 1: LOCO clock 1 0 0: IWDTC-dedicated clock (IWDTCCLK) 1 0 1: Peripheral module clock B (PCLKB) 1 1 0: USB clock (UCLK) 1 1 1: External clock for the Ethernet-PHY (CLKOUT25M)	R/W
b5, b4	RCDS[1:0]	Measurement Reference Clock Frequency Division Ration Select	b5 b4 0 0: ×1/32 clock 0 1: ×1/128 clock 1 0: ×1/1024 clock 1 1: ×1/8192 clock	R/W
b7, b6	DFS[1:0]	Digital Filter Select	b7 b6 0 0: Digital filtering is disabled. 0 1: The sampling clock for the digital filter is the measurement target clock. 1 0: The sampling clock for the digital filter is the measurement target clock divided by 4. 1 1: The sampling clock for the digital filter is the measurement target clock divided by 16.	R/W

Note: Set the CACR2 register when the CACR0.CFME bit is 0.

RPS Bit (Reference Signal Select)

This bit selects whether to use the CACREF pin input or an internal clock (internally generated signal) as the reference signal.

RSCS[2:0]Bits (Measurement Reference Clock Select)

These bits select the clock source for generating the measurement reference clock.

RCDS[1:0]Bits (Measurement Reference Clock Frequency Division Ration Select)

These bits select the frequency division ratio of the measurement reference clock.

DFS[1:0]Bits (Digital Filter Select)

The setting of these bits enables or disables the digital filter and selects its sampling clock.

10.2.4 CAC Interrupt Request Enable Register (CAICR)

Address(es): CAC.CAICR 0008 B003h

b7	b6	b5	b4	b3	b2	b1	b0
—	OVFFC L	MENDF CL	FERRF CL	—	OVFIE	MENDI E	FERRI E
Value after reset:	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	FERRIE	Frequency Error Interrupt Request Enable	0: Frequency error interrupt request is disabled. 1: Frequency error interrupt request is enabled.	R/W
b1	MENDIE	Measurement End Interrupt Request Enable	0: Measurement end interrupt request is disabled. 1: Measurement end interrupt request is enabled.	R/W
b2	OVFIE	Overflow Interrupt Request Enable	0: Overflow interrupt request is disabled. 1: Overflow interrupt request is enabled.	R/W
b3	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b4	FERRFCL	FERRF Clear	When 1 is written to this bit, the CASTR.FERRF flag is cleared. This bit is read as 0.	R/W
b5	MENDFCL	MENDF Clear	When 1 is written to this bit, the CASTR.MENDF flag is cleared. This bit is read as 0.	R/W
b6	OVFFCL	OVFF Clear	When 1 is written to this bit, the CASTR.OVFF flag is cleared. This bit is read as 0.	R/W
b7	—	Reserved	This bit is read as 0. The write value should be 0.	R/W

FERRIE Bit (Frequency Error Interrupt Request Enable)

This bit specifies whether the frequency error interrupt request is enabled or disabled.

MENDIE Bit (Measurement End Interrupt Request Enable)

This bit specifies whether the measurement end interrupt request is enabled or disabled.

OVFIE Bit (Overflow Interrupt Request Enable)

This bit specifies whether the overflow interrupt request is enabled or disabled.

FERRFCL Bit (FERRF Clear)

Setting this bit to 1 clears the CASTR.FERRF flag.

MENDFCL Bit (MENDF Clear)

Setting this bit to 1 clears the CASTR.MENDF flag.

OVFFCL Bit (OVFF Clear)

Setting this bit to 1 clears the CASTR.OVFF flag.

10.2.5 CAC Status Register (CASTR)

Address(es): CAC.CASTR 0008 B004h

b7	b6	b5	b4	b3	b2	b1	b0
—	—	—	—	—	OVFF	MENDF	FERRF
0	0	0	0	0	0	0	0

Value after reset:

Bit	Symbol	Bit Name	Description	R/W
b0	FERRF	Frequency Error Flag	0: The clock frequency is within the range corresponding to the settings. 1: The clock frequency has deviated beyond the range corresponding to the settings (frequency error).	R
b1	MENDF	Measurement End Flag	0: Measurement is in progress. 1: Measurement has ended.	R
b2	OVFF	Overflow Flag	0: The counter has not overflowed. 1: The counter has overflowed.	R
b7 to b3	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

FERRF Flag (Frequency Error Flag)

This flag indicates deviation of the clock frequency from the set value (frequency error).

[Setting condition]

- The clock frequency is outside of the setting range.

[Clearing condition]

- 1 is written to the CAICR.FERRFCL bit.

MENDF Flag (Measurement End Flag)

This flag indicates the end of measurement.

[Setting condition]

- Measurement has finished.

[Clearing condition]

- 1 is written to the CAICR.MENDFCL bit.

OVFF Flag (Overflow Flag)

This flag indicates that the counter has overflowed.

[Setting condition]

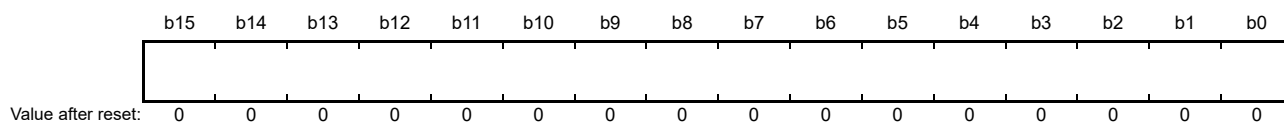
- The counter has overflowed.

[Clearing condition]

- 1 is written to the CAICR.OVFFCL bit.

10.2.6 CAC Upper-Limit Value Setting Register (CAULVR)

Address(es): CAC.CAULVR 0008 B006h



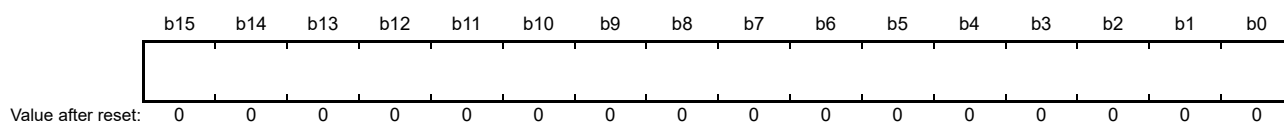
The CAULVR register is a 16-bit readable/writable register that specifies the upper-limit value of the counter used for measuring the frequency. When the frequency rises above the value specified in this register, a frequency error is detected.

Write to this register when the CACR0.CFME bit is 0.

The counter value held in the CACNTBR register can vary with the difference between the phases of the digital filter and edge-detection circuit on the one hand and the signal on the CACREF pin on the other, so ensure that this setting allows an adequate margin.

10.2.7 CAC Lower-Limit Value Setting Register (CALLVR)

Address(es): CAC.CALLVR 0008 B008h



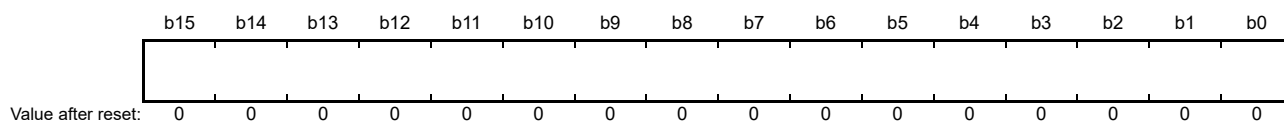
The CALLVR register is a 16-bit readable/writable register that specifies the lower-limit value of the counter used for measuring the frequency. When the frequency falls below the value specified in this register, a frequency error is detected.

Write to this register when the CACR0.CFME bit is 0.

The counter value held in the CACNTBR register can vary with the difference between the phases of the digital filter and edge-detection circuit on the one hand and the signal on the CACREF pin on the other, so ensure that this setting allows an adequate margin.

10.2.8 CAC Counter Buffer Register (CACNTBR)

Address(es): CAC.CACNTBR 0008 B00Ah



The CACNTBR register is a 16-bit read-only register that retains the counter value at the time a valid reference signal edge is input.

10.3 Operation

10.3.1 Measuring Clock Frequency

The clock frequency accuracy measurement circuit measures the clock frequency using the CACREF pin input or the internal clock as a reference. Figure 10.2 shows an operating example of the clock frequency accuracy measurement circuit.

The clock frequency accuracy measurement circuit operates as shown below when measuring the clock frequency.

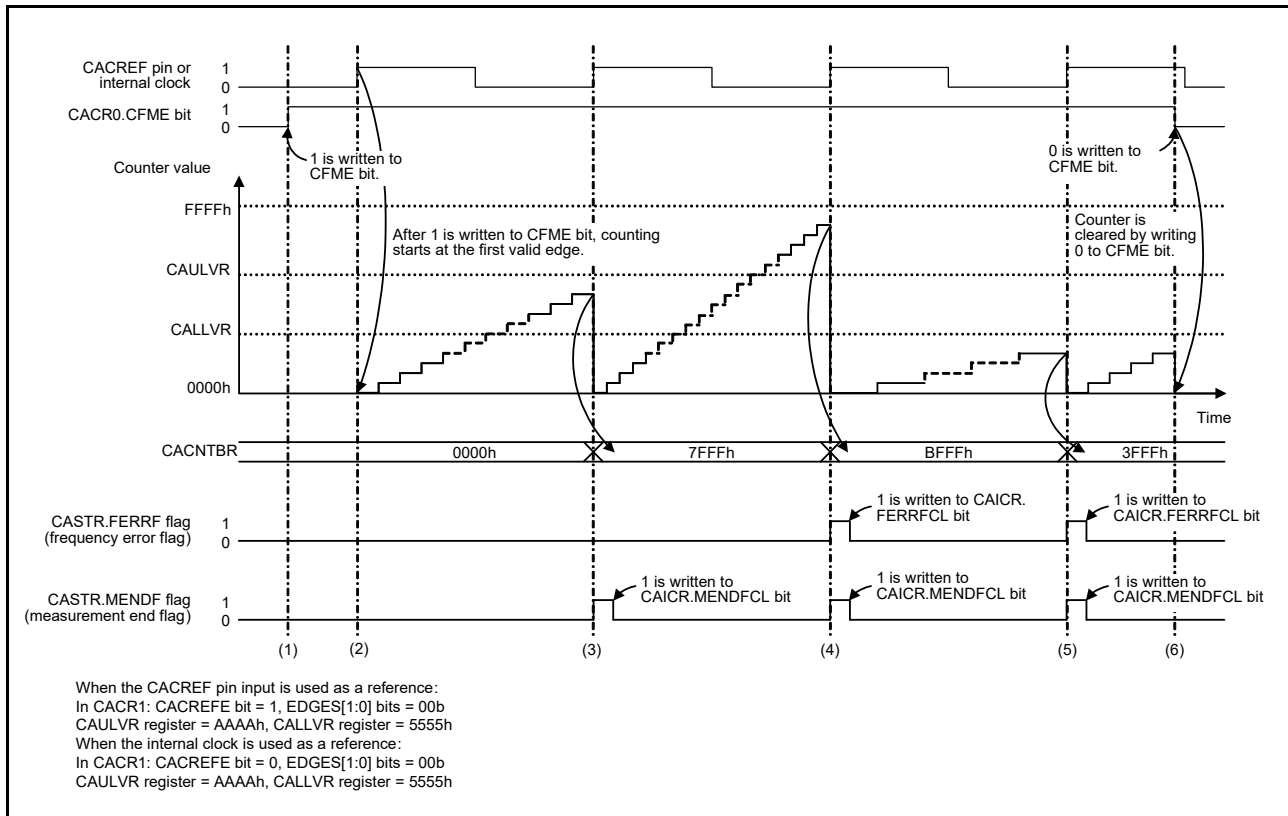


Figure 10.2 Operating Example of Clock Frequency Accuracy Measurement Circuit

- (1) When the CACREF pin input is used as a reference (the CACR1.CACREFE bit = 1), clock frequency measurement is enabled by writing 1 to the CACR0.CFME bit while the CACR2.RPS bit is 0 and the CACR1.CACREFE bit is 1. On the other hand, when the internal clock is used as a reference (the CACR1.CACREFE bit = 0), clock frequency measurement is enabled by writing 1 to the CACR0.CFME bit while the CACR2.RPS bit is 1.
- (2) When the CACREF pin input is used as a reference, the timer starts up-counting if the valid edge selected by the CACR1.EDGES[1:0] bits is input from the CACREF pin after 1 is written to the CFME bit. The valid edge is a rising edge (the CACR1.EDGES[1:0] bits = 00b) in Figure 10.2.
When the internal clock is used as a reference, the timer starts up-counting if the valid edge selected by the CACR1.EDGES[1:0] bits is input based on the clock source selected by the CACR2.RSCS[2:0] bits after 1 is written to the CFME bit. The valid edge is a rising edge (the CACR1.EDGES[1:0] bits = 00b) in Figure 10.2.
- (3) When the next valid edge is input, the counter value is transferred in the CACNTBR register and compared with the values of the CAULVR and CALLVR registers. If the formula $CALLVR \leq CACNTBR \leq CAULVR$ is satisfied, only the CASTER.MENDF flag is set to 1 because the clock frequency is correct. If the CAICR.MENDIE bit is 1, a measurement end interrupt is generated.
- (4) When the next valid edge is input, the counter value is transferred in the CACNTBR register and compared with the values of the CAULVR and CALLVR registers. In the case of $CACNTBR > CAULVR$, the CASTER.FERRF flag is set to 1 because the clock frequency is erroneous. If the CAICR.FERRIE bit is 1, a frequency error interrupt is

generated. Also, the CASTR.MENDF flag is set to 1. If the CAICR.MENDIE bit is 1, a measurement end interrupt is generated.

- (5) When the next valid edge is input, the counter value is transferred in the CACNTBR register and compared with the values of the CAULVR and CALLVR registers. In the case of $CACNTBR < CALLVR$, the CASTR.FERRF flag is set to 1 because the clock frequency is erroneous. If the CAICR.FERRIE bit is 1, a frequency error interrupt is generated. Also, the CASTR.MENDF flag is set to 1. If the CAICR.MENDIE bit is 1, a measurement end interrupt is generated.
- (6) While the CACR0.CFME bit is 1, the counter value is transferred in the CACNTBR register and compared with the values of the CAULVR and CALLVR registers every time a valid edge is input. Writing 0 to the CACR0.CFME bit clears the counter and stops up-counting.

10.3.2 Digital Filtering of Signals on the CACREF Pin

The CACREF pin has a digital filter. Levels on the target pin for sampling are conveyed to the internal circuitry after matching three consecutive times at the selected sampling interval and the same level continues to be conveyed internally until the level on the pin again matches three consecutive times.

Enabling and disabling of the digital filter and its sampling clock are selectable.

The counter value transferred in the CACNTBR register may be in error by up to one cycle of the sampling clock due to the difference between the phases of the digital filter and the signal input to the CACREF pin.

When a frequency dividing clock is selected as a count source clock, the counter value error is obtained by the following formula:

$$\text{Counter value error} = (\text{One cycle of the count source clock}) / (\text{One cycle of the sampling clock})$$

10.4 Interrupt Requests

The CAC generates three types of interrupt request: frequency error interrupt, measurement end interrupt, and overflow interrupt. When an interrupt source is generated, the corresponding status flag becomes 1. Table 10.3 lists details on the interrupt requests of the clock frequency accuracy measurement circuit.

Table 10.3 Interrupt Requests of Clock Frequency Accuracy Measurement Circuit

Interrupt Request	Interrupt Enable Bit	Status Flag	Interrupt Source
Frequency error interrupt	CAICR.FERRIE	CASTR.FERRF	The result of comparing CACNTBR to CAULVR and CALLVR is either $CACNTBR > CAULVR$ or $CACNTBR < CALLVR$.
Measurement end interrupt	CAICR.MENDIE	CASTR.MENDF	A valid edge is input from the CACREF pin. Note however that a measurement end interrupt does not occur at the first valid edge after writing 1 to the CACR0.CFME bit.
Overflow interrupt	CAICR.OVFIE	CASTR.OVFF	The counter has overflowed.

10.5 Usage Notes

10.5.1 Module Stop Function Setting

CAC operation can be disabled or enabled using module stop control register C (MSTPCRC). The initial setting is for the CAC to be halted. Register access is enabled by releasing the module stop state. For details, refer to **section 11, Low Power Consumption**.